TrailSense Multilateration Library: Output Reference

July 12, 2026

CONTENTS

TrailSense Multilateration Library: Output Reference

The position output contract: coordinate frame and units, accuracy, confidence, the error ellipse semantics, and the CLI JSON form. Applies to both the C API output and the CLI output. Part of the TrailSense Multilateration Library documentation set.

Output Format: The Position Contract

Every field emitted per position: frame, units, accuracy, confidence, error ellipse. Canonical home for error-ellipse semantics.

Two output forms, same underlying values:

- C struct `ts_position_t`, written by `ts_triangulate` into your `out` buffer.
- JSON per position from the `trailsense-triangulate` CLI, adding two derived fields (`signal_type`, `fingerprint_hash`).

Coordinate frame and units

Geographic lat/lon, decimal degrees.

- Datum: WGS 84 assumed. No datum transform; output is in the same geodetic frame as the surveyed anchors you supply (WGS 84 anchors in means WGS 84 out).
- Axis order: emits `lat` THEN `lon`. GeoJSON and many geospatial consumers order `[longitude, latitude]`; a `lon, lat` consumer requires swapping. Not swapping puts every point in the wrong hemisphere or ocean.
- Precision: CLI renders `lat`/`lon` at 9 decimal places, far below any RF fix's physical accuracy; trailing digits are numerical, not real precision.

accuracy_m

Weighted CEP (circular error probable) estimate, meters.

- 3-or-more-node fix: weighted mean absolute range residual over surviving measurements after robust outlier handling.
- 2-node fix: set equal to `error_semi_major_m` (larger axis).

Caveat: the percentile `accuracy_m` represents is NOT formally specified. “CEP” conventionally means a 50 percent radius, but this is a residual-derived estimate, not a calibrated statistical radius, so assume no specific confidence percentile. Treat as a relative quality indicator (smaller is better), not a guaranteed containment radius. Called out honestly rather than assigned an invented percentile.

confidence

Discrete geometry tier, not a probability or percentage.

Value	Anchor count	Meaning
`40`	exactly 2	Approximate bilateration. Unresolvable off the baseline; fix on the baseline with a mirror ambiguity. Lowest tier.
`70`	exactly 3	Full multilateration, minimum over-determination for a 2D fix.
`85`	4 or more	Full multilateration with additional redundancy. Highest tier.

Reflects only how many distinct anchors participated (which solver ran); does not encode measured position error. Do not read `confidence / 100` as a probability.

measurement_count

Participating anchors: distinct enrolled, resolvable anchors that saw the device within the batch. NOT the post-outlier survivor count; the robust 3+ solver may internally down-weight or reject a measurement, but this reports the count that entered the solve. Always 2 for a 2-node fix.

peak_rssi

Strongest (least negative) RSSI, dBm, among participating observations for this device and band.

Error ellipse

Fields `error_semi_major_m`, `error_semi_minor_m`, `error_major_axis_deg`. Meaning depends on anchor count; the 2-node case is subtle.

ERROR_MAJOR_AXIS_DEG

Bearing of the semi-major axis: 0 = North, clockwise, folded into `[0, 180)` (an axis has no head or tail).

2-NODE FIX (CONFIDENCE 40): MIXED SEMANTICS

Not a single Gaussian; the two axes mean different things, and one is not a statistical radius at all:

- Along-baseline axis: a genuine roughly 1-sigma statistical range uncertainty, from propagating RSSI-to-range noise onto the along-baseline coordinate. Scales as roughly `r / d` (target range over baseline length), unbounded as baseline `d` approaches 0. Only the old quadratic GDOP blow-up (`1 / det^2`) was removed; a linear `1 / d` growth remains. A minimum-baseline gate (default 5 meters) drops near-coincident 2-node pairs before this term explodes.
- Cross-baseline axis: a DETERMINISTIC mirror-ambiguity bound, the larger of the mirror half-separation `h` and the geometric-inconsistency gap. With only two range circles the true target sits at plus-or-minus `h` off the baseline while the estimate is placed on the baseline, so this is a hard bound on perpendicular displacement. NOT a 1-sigma value, NOT a confidence radius. Consumers MUST NOT treat the cross axis as a confidence radius.

When the range circles are inconsistent with the baseline (non-intersecting, `d > r1 + r2`, or nested, `d < |r1 - r2|`), the along-track clamp would drive `h` toward 0 and report a falsely tight cross axis on broken geometry. To avoid that, the cross axis is raised to the gap: `sigma_cross = max(h, gap)` with `gap = max(0, d - (r1 + r2), |r1 - r2| - d)`. On consistent geometry `gap == 0`, so `sigma_cross == h`, unchanged.

Larger axis becomes `error_semi_major_m`, smaller becomes `error_semi_minor_m`, `accuracy_m == error_semi_major_m`. Because the axes carry different meanings, do not interpret this ellipse as a single elliptical confidence region.

3-OR-MORE-NODE FIX: ISOTROPIC PLACEHOLDER

- `error_semi_major_m == error_semi_minor_m == accuracy_m`
- `error_major_axis_deg == 0`

Full 3-plus-node covariance is not computed; no directional error information on this path. Circle radius is `accuracy_m`.

No timestamp field

Output has NO timestamp field. Neither `ts_position_t` nor the JSON position object carries a time. The library is stateless and clockless; it never subdivides the batch by time and never stamps the result.

The consumer must attach its own time: record the receive or observation time per batch and associate it with the returned positions. Staleness checks, track filtering, and time-ordered storage are the consumer’s responsibility. See the Edge cases section of the Usage Guide.

JSON form (CLI output)

CLI wraps positions in a top-level object:

```
{"positions":[ ... ]}
```

Each element carries every `ts_position_t` field plus two derived fields:

- `signal_type`: `"wifi"`, `"bluetooth"`, or `"cellular"` for band 0, 1, 2.
- `fingerprint_hash`: band prefix (`w_`, `b_`, `c_`) + the 8 lowercase hex chars of `mac_hash`, e.g. `"w_deadbeef"`.

`mac_hash` renders as 8 lowercase hex chars. Fixed field order: `mac_hash`, `band`, `signal_type`, `fingerprint_hash`, `lat`, `lon`, `accuracy_m`, `measurement_count`, `confidence`, `peak_rssi`, `error_semi_major_m`, `error_semi_minor_m`, `error_major_axis_deg`.

Worked example (annotated)

`trailsense-triangulate` on `examples/sample-input.json` produces `examples/sample-output.json`:

```
{
  "positions": [
    {
      "mac_hash": "deadbeef",
      "band": 0,
      "signal_type": "wifi",
      "fingerprint_hash": "w_deadbeef",
      "lat": 44.999675310,
      "lon": -93.000735291,
      "accuracy_m": 17.522416693,
      "measurement_count": 3,
      "confidence": 70,
      "peak_rssi": -55,
      "error_semi_major_m": 17.522416693,
      "error_semi_minor_m": 17.522416693,
      "error_major_axis_deg": 0.000000000
    },
    {
      "mac_hash": "cafef00d",
      "band": 1,
      "signal_type": "bluetooth",
      "fingerprint_hash": "b_cafef00d",
      "lat": 45.000000000,
      "lon": -93.000000000,
      "accuracy_m": 984.562871623,
      "measurement_count": 2,
      "confidence": 40,
      "peak_rssi": -70,
      "error_semi_major_m": 984.562871623,
      "error_semi_minor_m": 79.430000000,
      "error_major_axis_deg": 0.000000000
    }
  ]
}
```


Position 1 `deadbeef`, 3-anchor Wi-Fi fix: `signal_type`/`fingerprint_hash` from `band` 0; `lat` before `lon` (swap for GeoJSON); `measurement_count` 3 + `confidence` 70 = full multilateration; isotropic placeholder ellipse (`error_semi_major_m == error_semi_minor_m == accuracy_m == 17.522416693`, `error_major_axis_deg == 0`).

Position 2 `cafef00d`, 2-anchor BLE fix: `measurement_count` 2 + `confidence` 40 = approximate bilateration; `accuracy_m` 984.56 m == `error_semi_major_m`, the along-baseline roughly 1-sigma range uncertainty, large here because the anchors are only about 10 m apart while the device is roughly 80 m away (large `r / d`), not a degenerate blow-up; `error_semi_minor_m` 79.43 m is the cross-baseline mirror-ambiguity bound (off-baseline half-separation), a hard perpendicular-displacement bound, NOT a 1-sigma or confidence radius, do not draw as a Gaussian error circle; `error_major_axis_deg` 0.000000000 is the baseline orientation folded into `[0, 180)` (vertically-aligned pair, major axis North).

Neither position carries a timestamp; the consumer attaches its own observation time.

